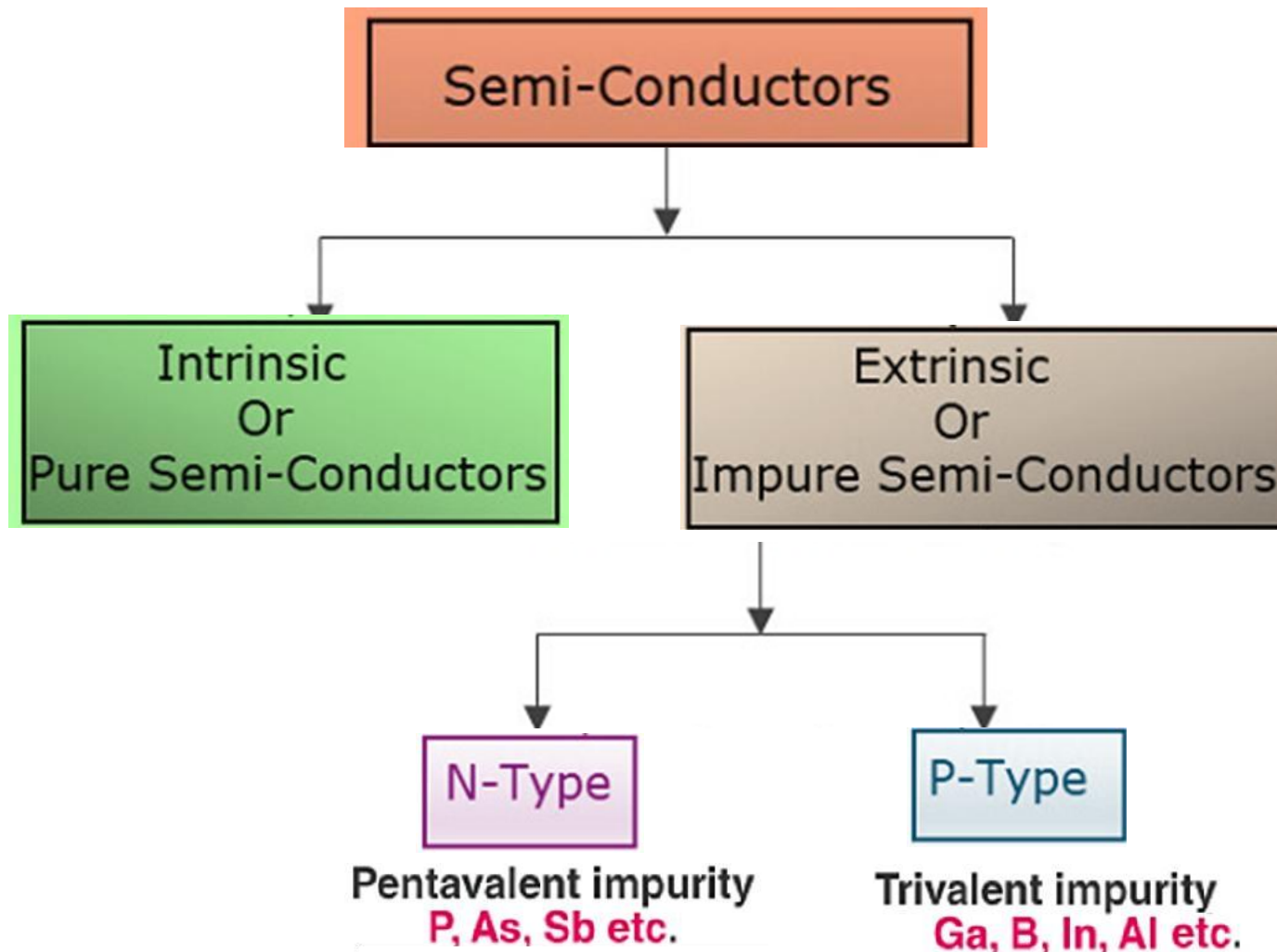


# **Electronics & Optoelectronic Devices**

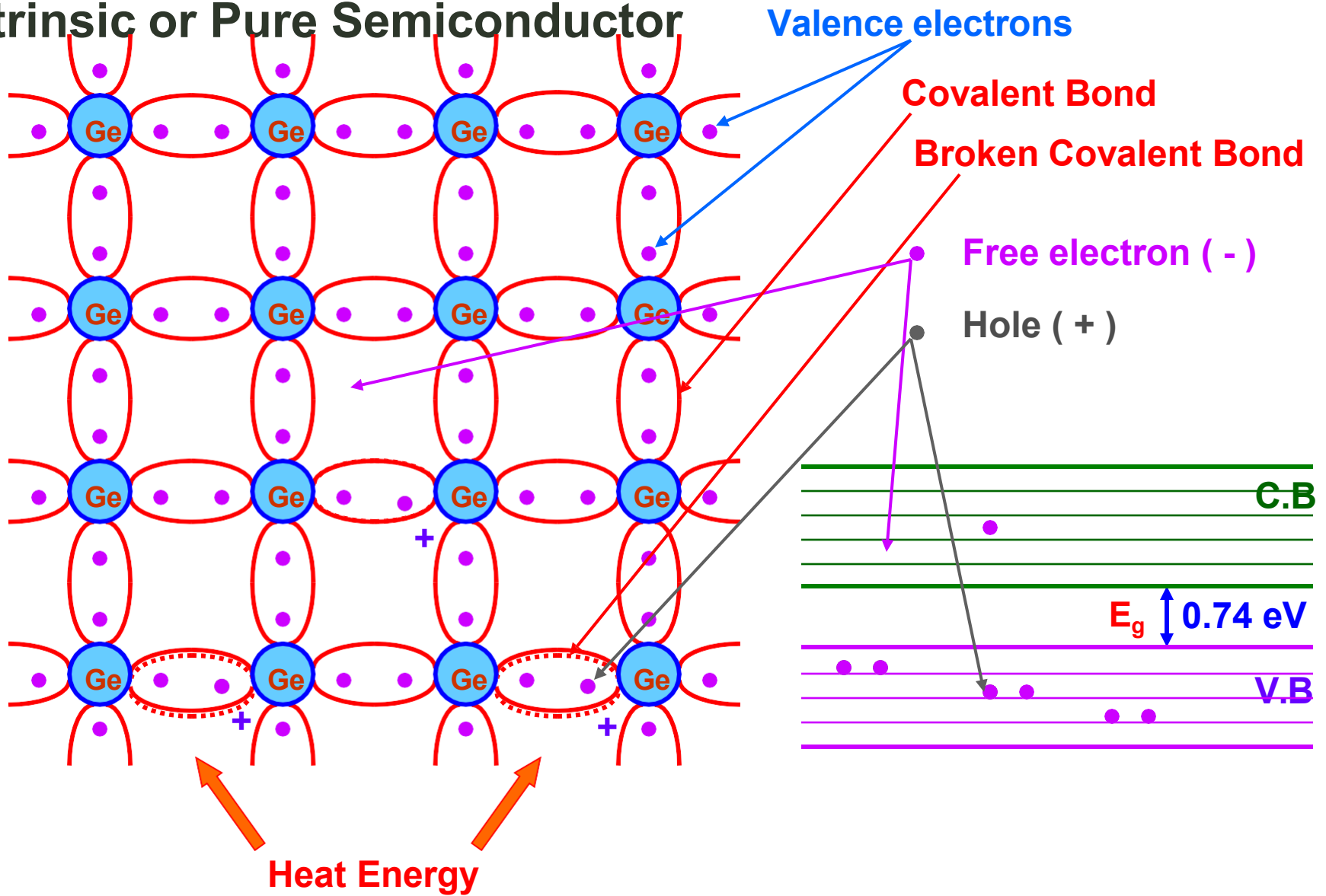
| <b>Module No.</b> | <b>Unit No.</b>                                 | <b>Details</b>                                                                                                                                                                                                                                                                                                                    | <b>Hrs.</b> | <b>CO</b>  |
|-------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------|
| <b>1</b>          | <b>Electronics &amp; Optoelectronic Devices</b> |                                                                                                                                                                                                                                                                                                                                   | <b>09</b>   | <b>CO1</b> |
|                   | <b>1.1</b>                                      | <b>Diodes and Transistors:</b> p-n junction at equilibrium and under bias, diode equation, rectifier diode, Zener diode and Schottky diode characteristics, transistor biasing and static characteristics in CB, CE and CC modes, operating points, transistor as a switch, MOS structure, charge storage and flash memory device |             |            |
|                   |                                                 | LEDs, laser diodes, photodetectors, solar cells, Organic LEDs, (OLEDs), Quantum Dot LEDs (QLEDs)                                                                                                                                                                                                                                  |             |            |

# Semiconductors

- ❖ Semiconductors are materials having energy band gap approximately 1eV i.e in between conductors and insulators.
- ❖ They have conductivity between conductors (generally metals) and nonconductors or insulators (such as most ceramics).
- ❖ Resistivity:  $10^{-5}$  to  $10^6 \Omega\text{m}$
- ❖ Conductivity:  $10^5$  to  $10^{-6}$  mho/m
- ❖ Temperature coefficient of resistance: Negative
- ❖ Current flow: Due to electrons and holes
- ❖ Semiconductors can be pure elements, such as silicon or germanium. They are called **Intrinsic semiconductors**.



# Intrinsic or Pure Semiconductor

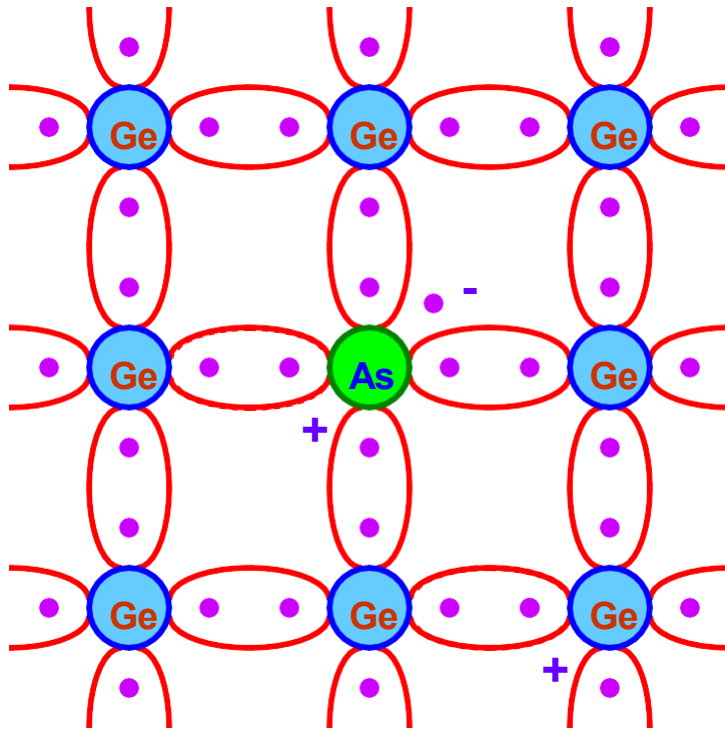


## Drawbacks of intrinsic or pure semiconductors

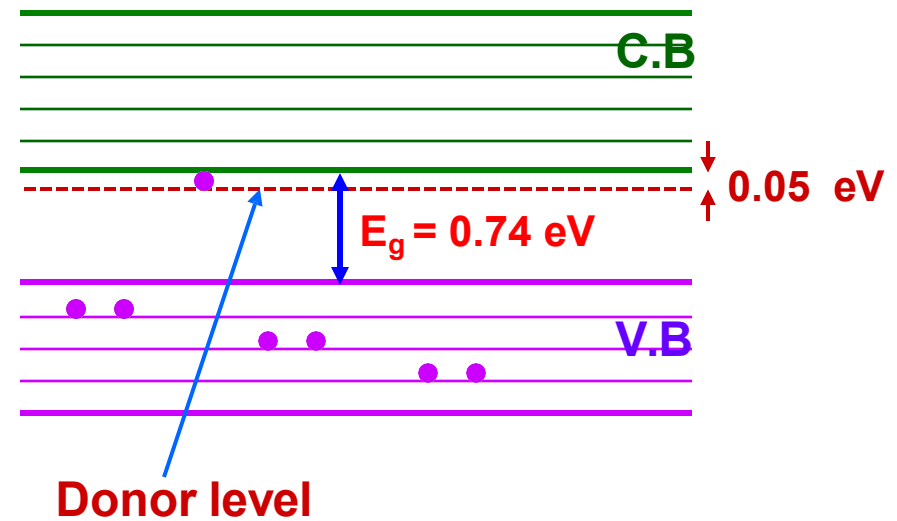
- ❖ A room temperature, it has a very small number current carriers (electrons and holes).
- ❖ Hence, its conductivity is low.
- ❖ Dependence on temperature only.

# DOPING IN SEMICONDUCTORS

- Semiconducting materials are very sensitive to impurities in the crystal lattice.
- The controlled addition of these impurities into pure semiconductor is known as **doping**.
- Allows the tuning of the electronic properties: technological applications.
- Introduction of dopants → 'extrinsic semiconductors'.
  - (a) Increase in the conductivity
  - (b) New intra-band, energy levels
  - (c) Generation of positive or negative charge carriers.



## N - Type Semiconductors

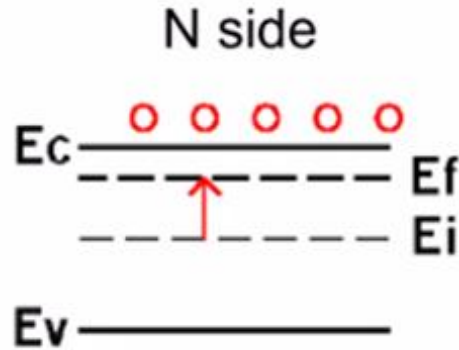


When a semiconductor such as Si or Ge is doped with a pentavalent impurity (Group V elements such as P, As or Sb), N - type semiconductor is formed.

When germanium (Ge) is doped with arsenic (As), the four valence electrons of As form covalent bonds with four Ge atoms and the fifth electron of As atom is loosely bound.



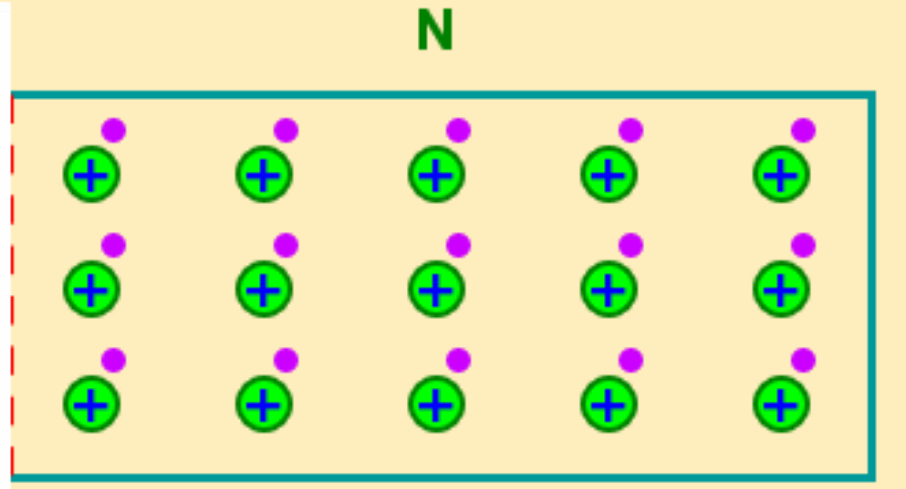
- The energy required to detach the fifth loosely bound electron is only of the order of 0.05 eV for germanium.
- Therefore, the energy state corresponding to the fifth electron is in the forbidden gap and slightly below the lower level of the conduction band.
- This energy level is called 'donor level'. The impurity atom is called N - type semiconductor is called donor-type semiconductor'.
- When intrinsic semiconductor is doped with donor impurities, not only does the number of electrons increase, but also the number of holes decreases.
- Therefore, in an N-type semiconductor, free electrons are the majority charge carriers and holes are the minority charge carriers.



$$n \approx N_d$$

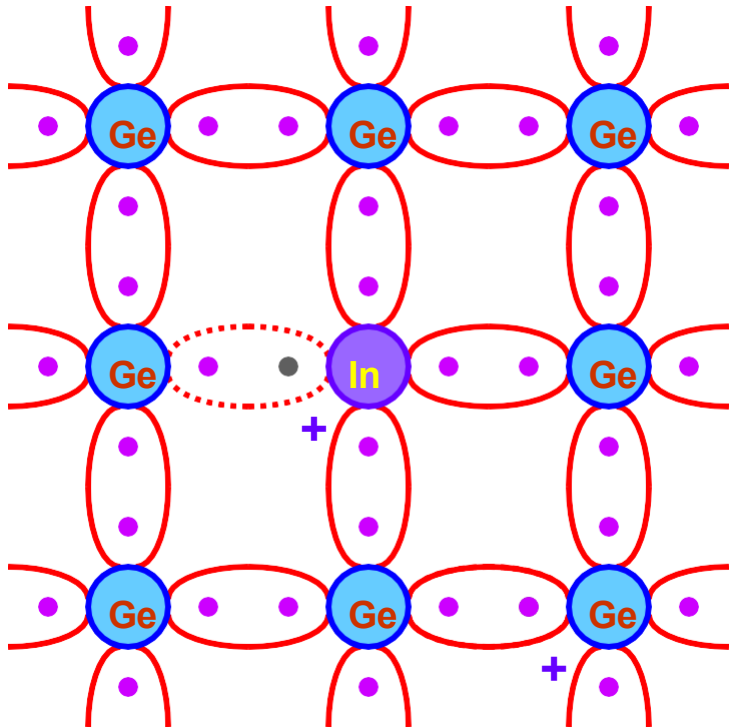
$$N_d = n_i e^{(E_F - E_i)/kT}$$

$$(E_F - E_i) = kT \ln \left( \frac{N_d}{n_i} \right)$$

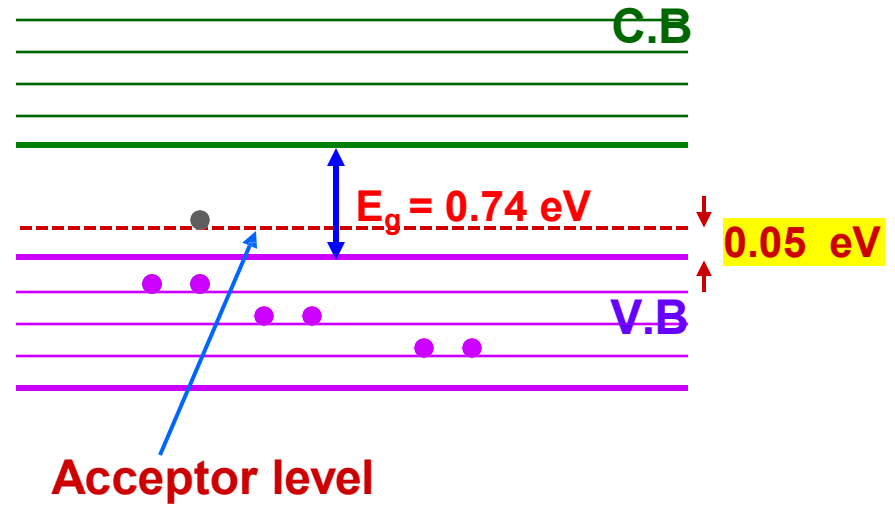


• Mobile Electron (Majority Carrier)

⊕ Immobile Positive Impurity Ion



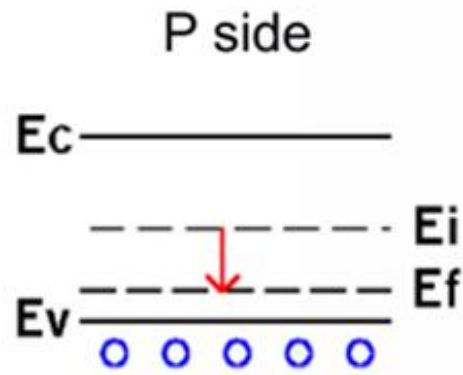
## P - Type Semiconductors



When a semiconductor such as Si or Ge is doped with a trivalent impurity (Group III elements such as In, B or Ga), P - type semiconductor is formed.

After doping, the three valence electrons of impurity form three covalent bonds with three Ge atoms. The vacancy that exists with the fourth covalent bond constitutes a **hole**.

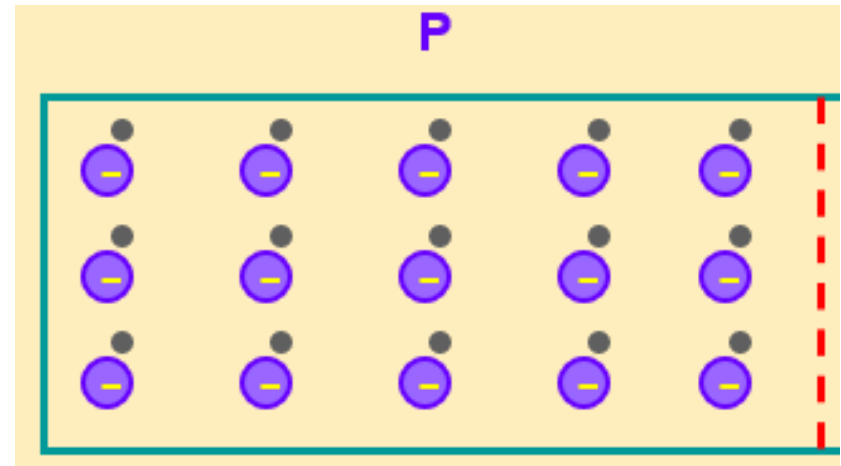
- ❖ The hole may be filled with an electron from neighboring atom, creating a hole in that position from where the electron jumped.
- ❖ Therefore, the trivalent impurity atom is called 'acceptor'.
- ❖ Since the hole is associated with a positive charge moving from one position to another, therefore, this type of semiconductor is called P - type semiconductor.
- ❖ The acceptor impurity produces an energy level just above the valence band known as 'acceptor level'.
- ❖ P - type semiconductor is called 'acceptor - type semiconductor'.
- ❖ In a P - type semiconductor, holes are the majority charge carriers and the electrons are the minority charge carriers.



$$p \approx N_a$$

$$N_a = n_i e^{(E_i - E_F)/kT}$$

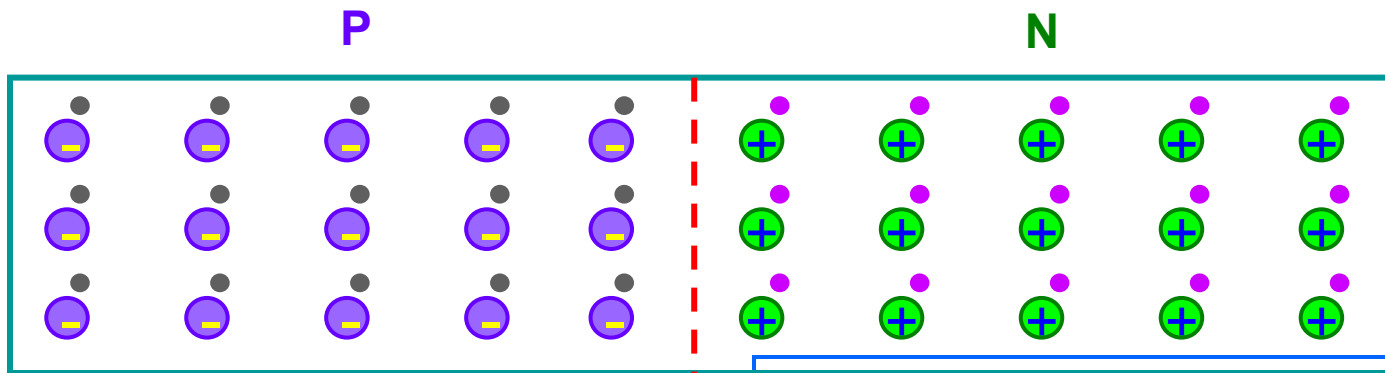
$$(E_i - E_F) = kT \ln \left( \frac{N_a}{n_i} \right)$$



- Mobile Hole (Majority Carrier)
- ⊖ Immobile Negative Impurity Ion

## PN Junction Diode:

When a P-type semiconductor is joined to a N-type semiconductor such that the crystal structure remains continuous at the boundary, the resulting arrangement is called a **PN junction diode** or a **semiconductor diode** or a **crystal diode**.



When a PN junction is formed, the P region has mobile holes (+) and immobile negatively charged ions.

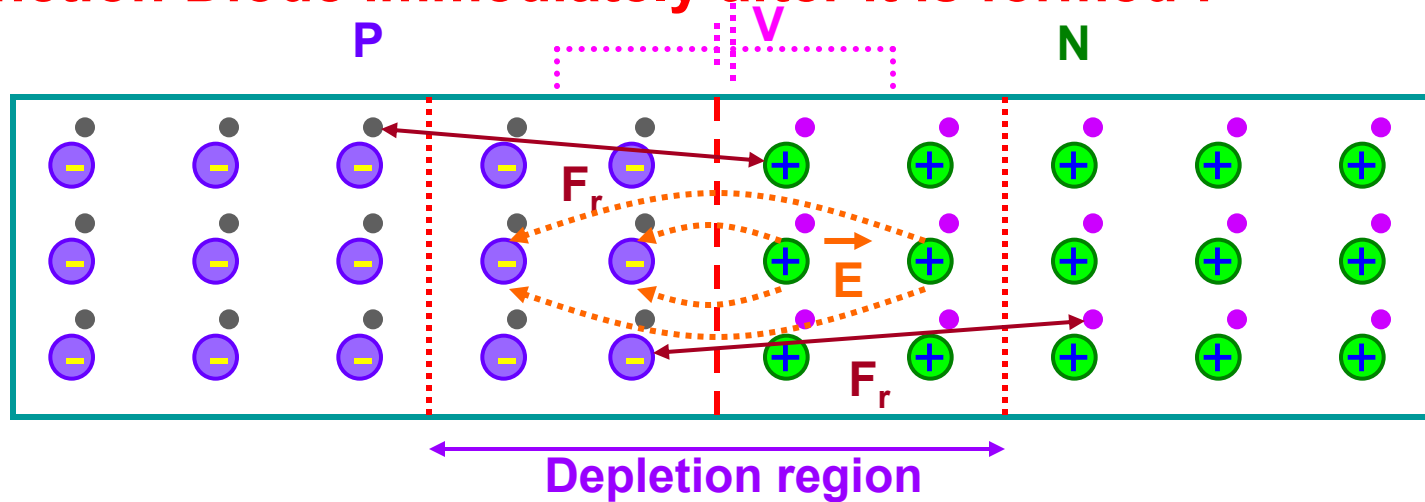
N region has mobile electrons (-) and immobile positively charged ions.

- Mobile Hole (Majority Carrier)
- Immobile Negative Impurity Ion
- Mobile Electron (Majority Carrier)
- Immobile Positive Impurity Ion

The whole arrangement is electrically neutral.

For simplicity, the minority charge carriers are not shown in the figure.

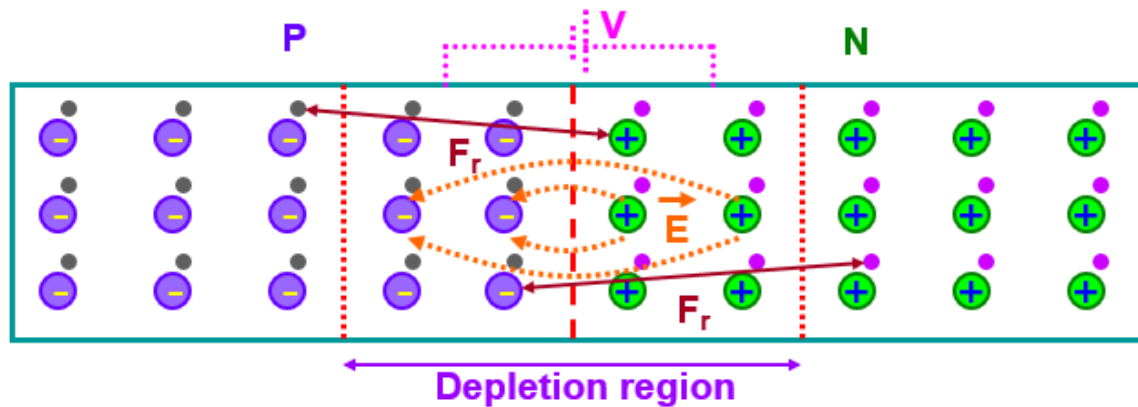
## PN Junction Diode immediately after it is formed :



After the PN junction diode is formed –

- i) Holes from P region diffuse into N region due to difference in concentration.
- ii) Free electrons from N region diffuse into P region due to the same reason.
- iii) Holes and free electrons combine near the junction.
- iv) Each recombination eliminates an electron and a hole.
- v) The uncompensated negative immobile ions in the P region do not allow any more free electrons to diffuse from N region.
- vi) The uncompensated positive immobile ions in the N region do not allow any more holes to diffuse from P region.

## How much is the Built-in Voltage?



$$qV_{bi} = (E_i - E_F)_{Left} + (E_F - E_i)_{Right}$$

$$V_{bi} = \frac{kT}{q} \ln\left(\frac{N_a}{n_i}\right) + \frac{kT}{q} \ln\left(\frac{N_d}{n_i}\right)$$

$$\Rightarrow V_{bi} = \frac{kT}{q} \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

$N_a$  acceptor level on the p side

$N_d$  donor level on the n side



## Special case: One-sided Junctions

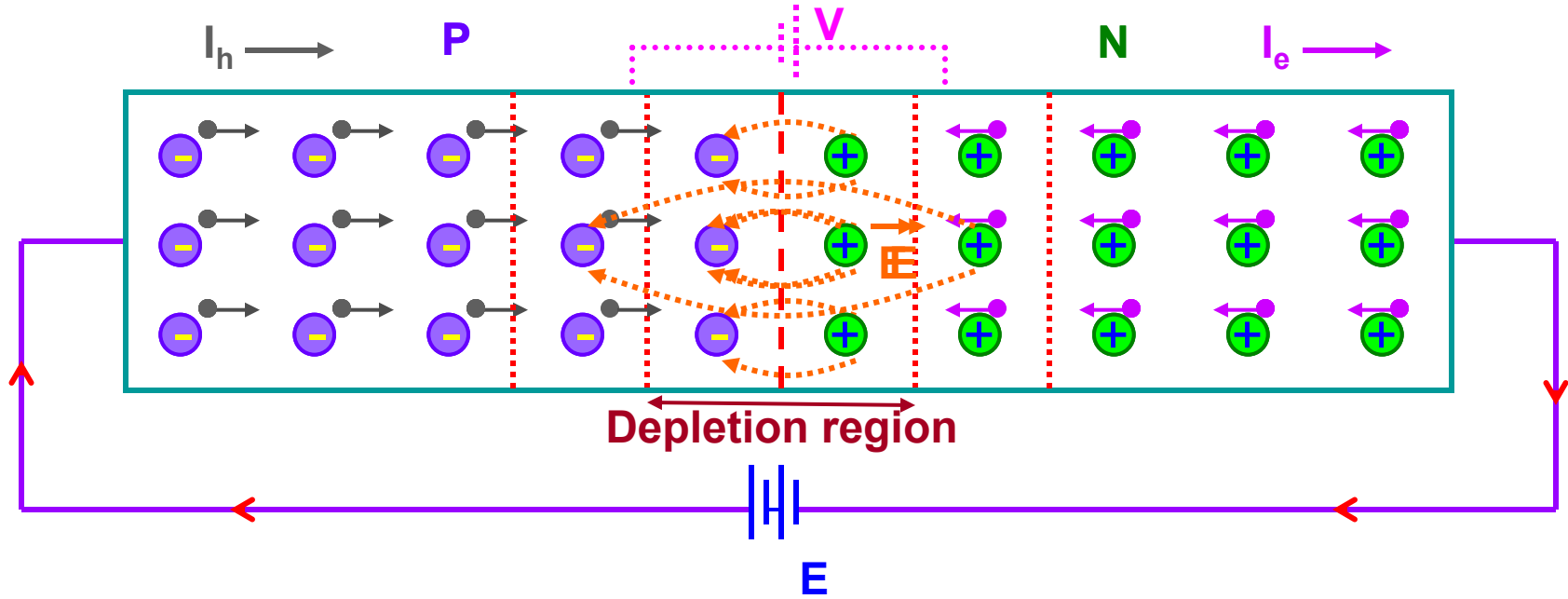
- One side very heavily doped so that Fermi level is at band edge.
- e.g. p<sup>+</sup>-n junction (heavy B implant into lightly doped substrate)

$$(E_i - E_F)_{\text{Left}} \approx E_i - E_V = E_G / 2$$

$$(E_F - E_i)_{\text{Right}} = kT \ln \left( \frac{N_D}{n_i} \right)$$

$$\Rightarrow V_{bi} = \frac{E_G}{2q} + \frac{kT}{q} \ln \left( \frac{N_d}{n_i} \right)$$

## Forward Bias:



When the positive terminal of the battery is connected to P-region and negative terminal is connected to N-region, then the PN junction diode is said to be forward-biased.

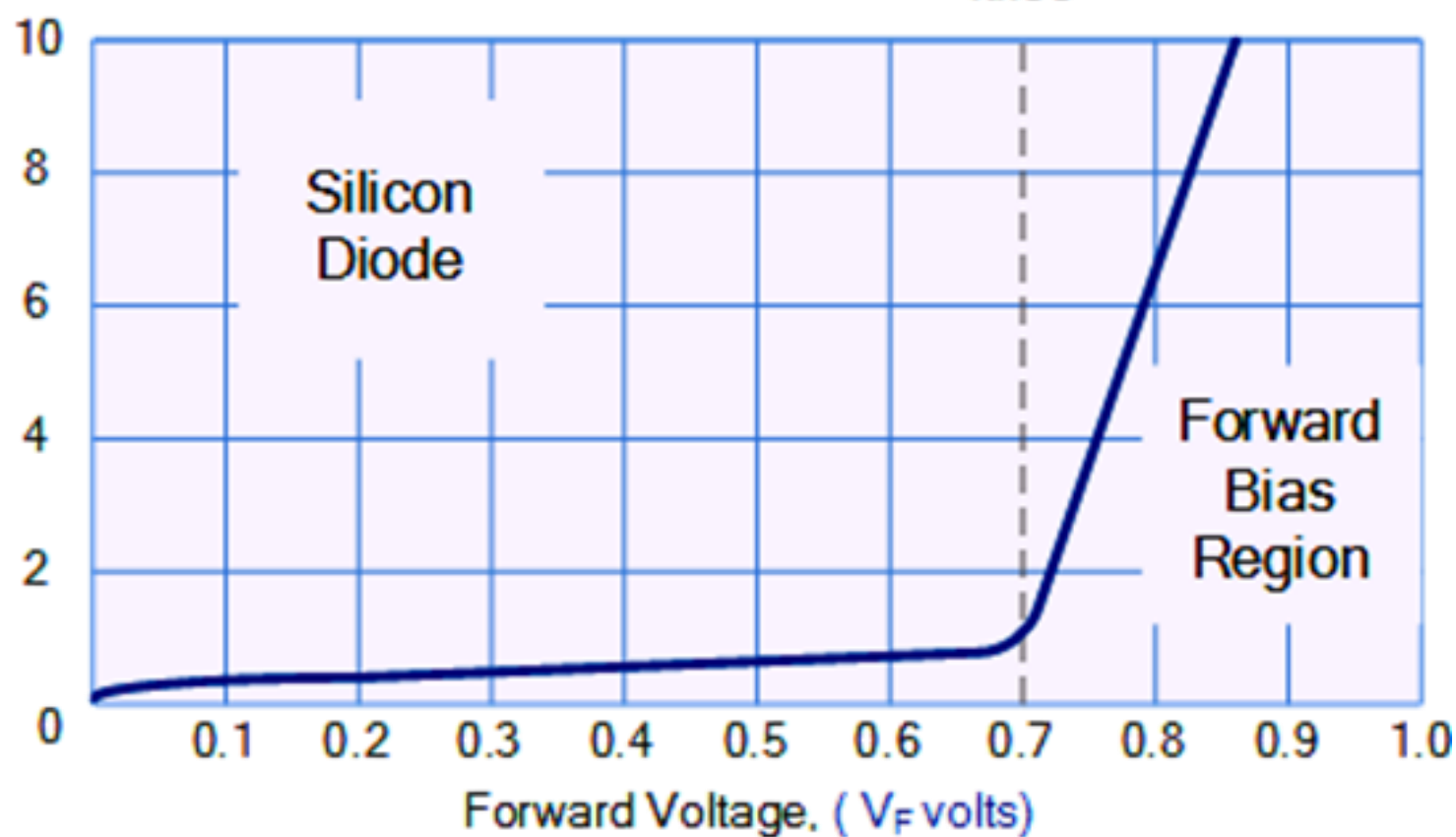
- i) Holes in P-region are repelled by +ve terminal of the battery and the free electrons are repelled by -ve terminal of the battery.
- ii) So, some holes and free electrons enter into the depletion region.
- iii) The potential barrier and the width of the depletion region decrease.
- iv) Therefore, a large number of majority carriers diffuse across the junction.
- v) Hole current and electronic current are in the same direction and add up.

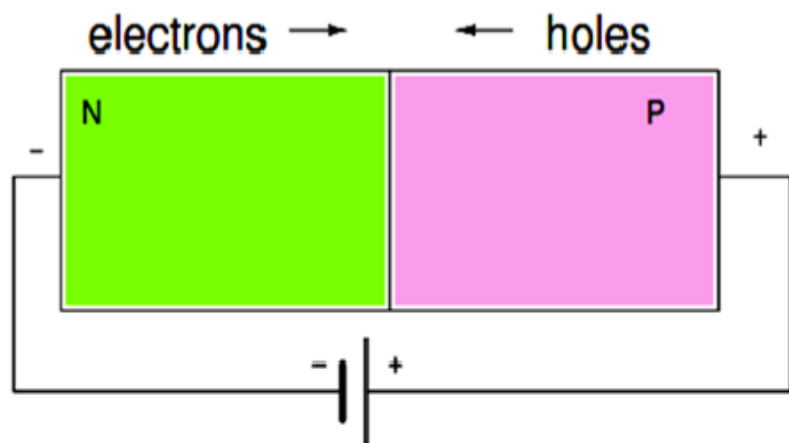
- v) Once they cross the junction, the holes in N-region and the electrons in P-region become minority carriers of charge and constitute minority current.
- vi) For each electron – hole recombination, an electron from the negative terminal of the battery enters the N-region and then drifts towards the junction.

In the P-region, near the positive terminal of the battery, an electron breaks covalent bond in the crystal and thus a hole is created. The hole drifts towards the junction and the electron enters the positive terminal of the battery.

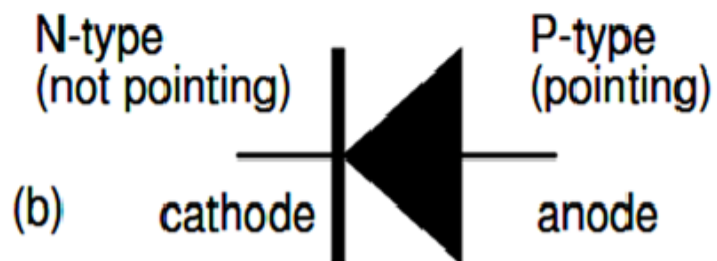
- vii) Thus, the current in the external circuit is due to movement of electrons, current in P-region is due to movement of holes and current in N-region is due to movement of electrons.
- viii) If the applied potential is increased, the potential barrier further decreases. As a result, a large number of majority carriers diffuse through the junction and a larger current flows.

Forward Current  
( $I_F$  mA)

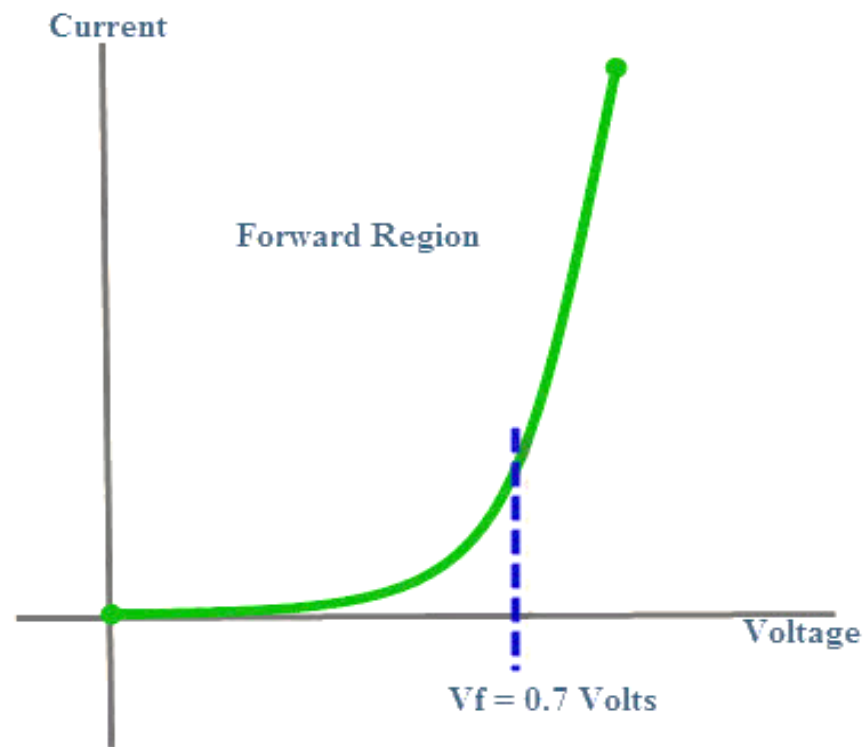




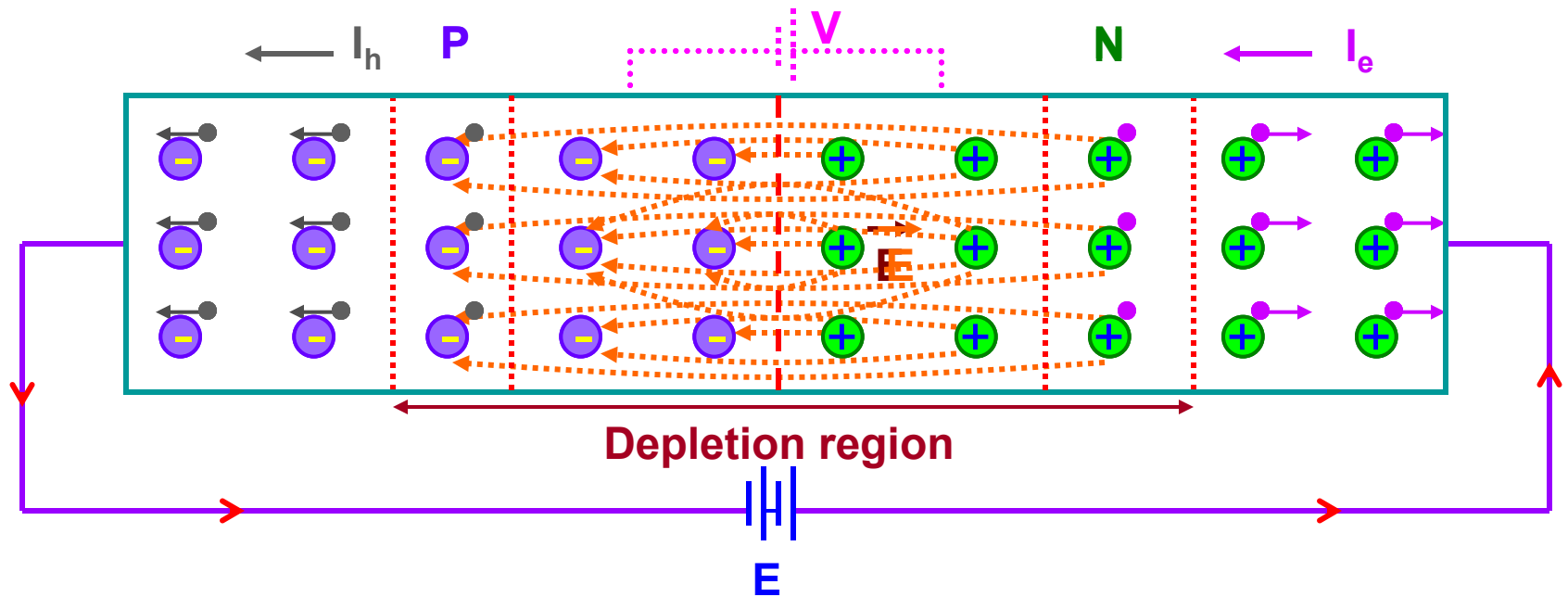
(a)



(b)



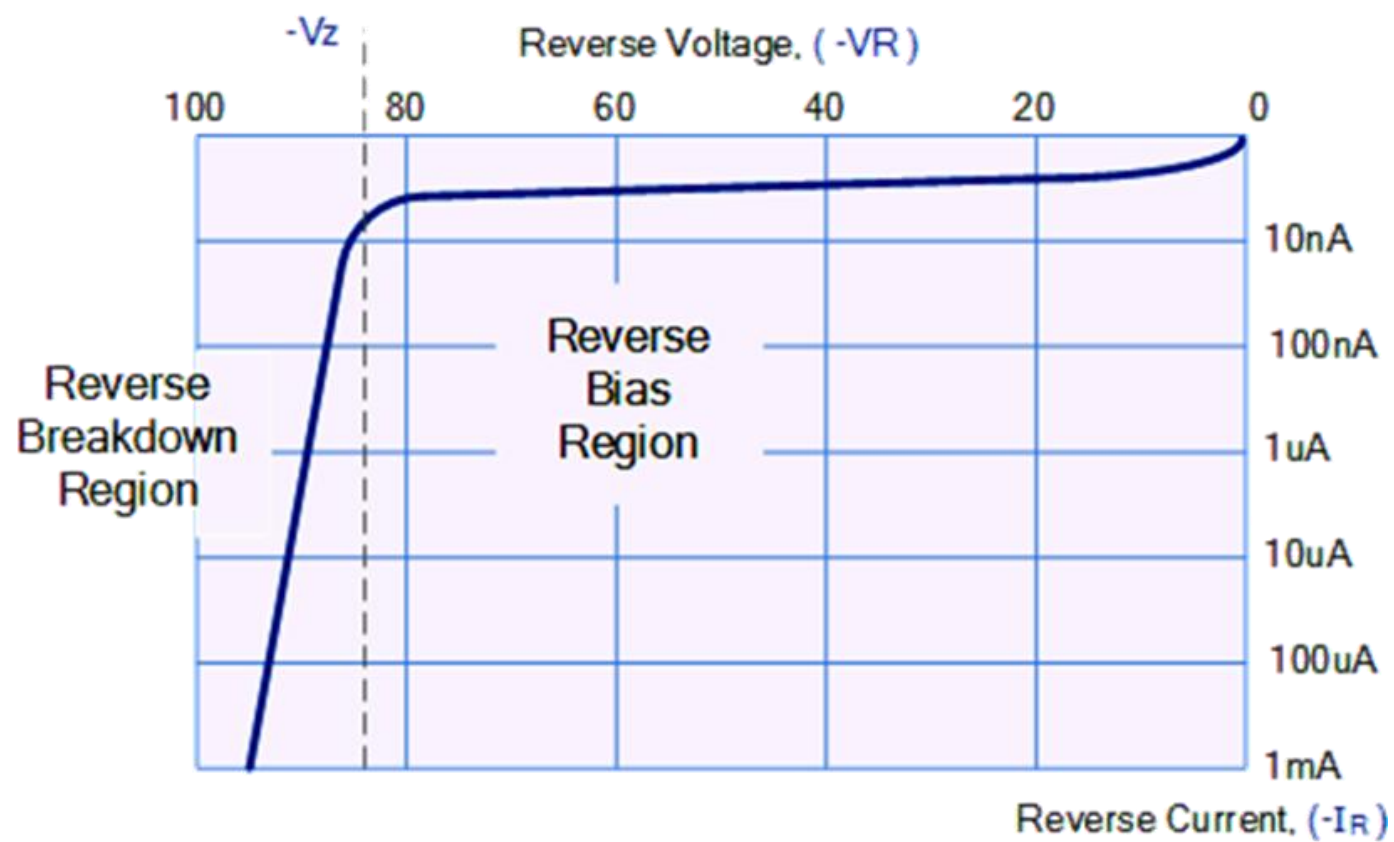
## Reverse Bias:



When the negative terminal of the battery is connected to P-region and positive terminal is connected to N-region, then the PN junction diode is said to be reverse-biased.

- i) Holes in P-region are attracted by -ve terminal of the battery and the free electrons are attracted by +ve terminal of the battery.
- ii) Thus, the majority carriers are pulled away from the junction.
- iii) The potential barrier and the width of the depletion region increase.
- iv) Therefore, it becomes more difficult for majority carriers diffuse across the junction.

- v) But the potential barrier helps the movement of the minority carriers. As soon as the minority carriers are generated, they are swept away by the potential barrier.
- vi) At a given temperature, the rate of generation of minority carriers is constant.
- vii) So, the resulting current is constant irrespective of the applied voltage. For this reason, this current is called 'reverse saturation current'.
- viii) Since the number of minority carriers is small, therefore, this current is small and is in the order of  $10^{-9}$  A in silicon diode and  $10^{-6}$  A in germanium diode.
- ix) The reverse – biased PN junction diode has an effective capacitance called 'transition or depletion capacitance'. P and N regions act as the plates of the capacitor and the depletion region acts as a dielectric medium.





# Ideal Diodes

The diode equation gives an expression for the current through a diode as a function of voltage. The *Ideal Diode Law*, expressed as:

$$I = I_0 \left( e^{\frac{qV}{kT}} - 1 \right)$$

where:

$I$  = the net current flowing through the diode;

$I_0$  = "dark saturation current", the diode leakage current density in the absence of light;

$V$  = applied voltage across the terminals of the diode;

$q$  = absolute value of electron charge ;

$k$  = Boltzmann's constant ; and

$T$  = absolute temperature (K)

At 300K,  $kT/q = 25.85$  mV, the "thermal voltage".

## Non-Ideal Diodes

For actual diodes, the expression becomes:

$$I = I_0 \left( e^{\frac{qV}{nkT}} - 1 \right)$$

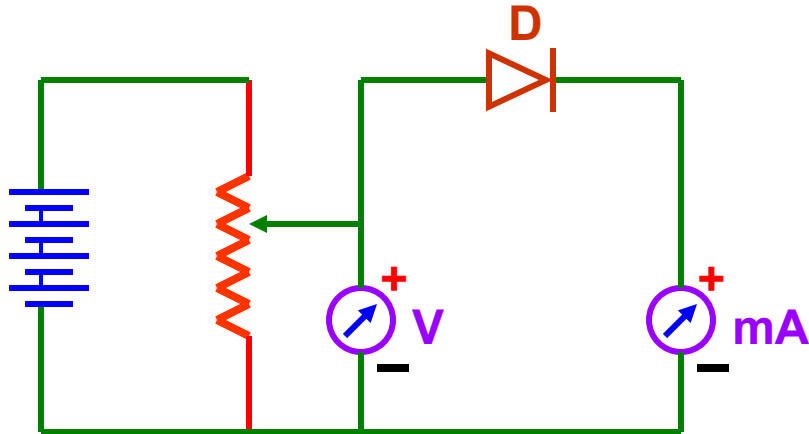
where:

$n$  = ideality factor, a number between 1 and 2 which typically increases as the current decreases.

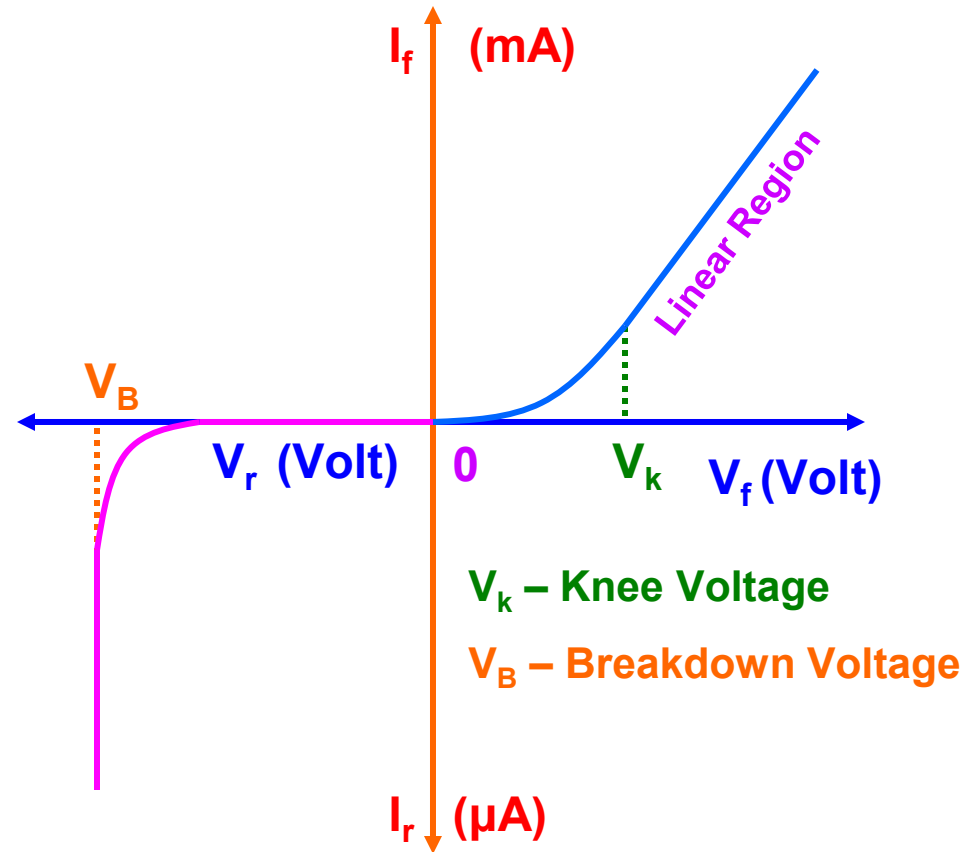
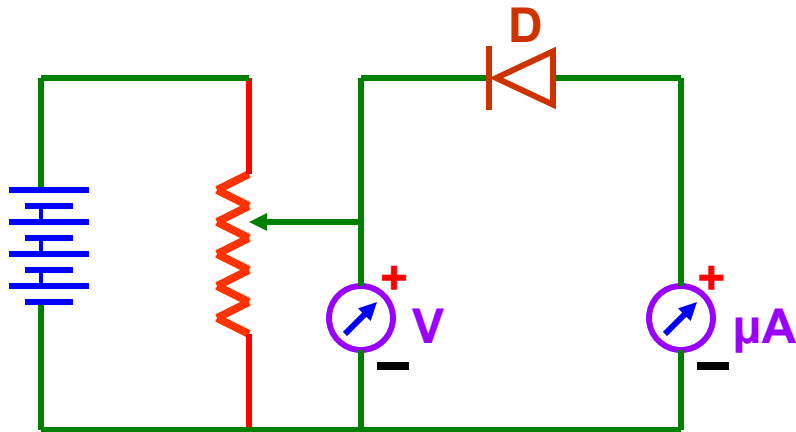
Reverse saturation current increases significantly with temperature because thermal energy boosts the generation of electron-hole pairs, creating more minority carriers to cross the junction; for silicon diodes roughly **doubles for every 10°C rise**, or increases by about 7% per degree Celsius, following the formula

**Formula:** This can be expressed as  $I_{s2} = I_{s1} \cdot 2^{(T_2 - T_1)/10}$ , where  $I_{s1}$  is current at  $T_1$  and  $I_{s2}$  is current at  $T_2$ .

## Diode Characteristics: Forward Bias:



## Reverse Bias:



## Resistance of a Diode:

i) Static or DC Resistance  $R_{d.c} = V / I$

ii) Dynamic or AC Resistance

$$R_{a.c} = \Delta V / \Delta I$$

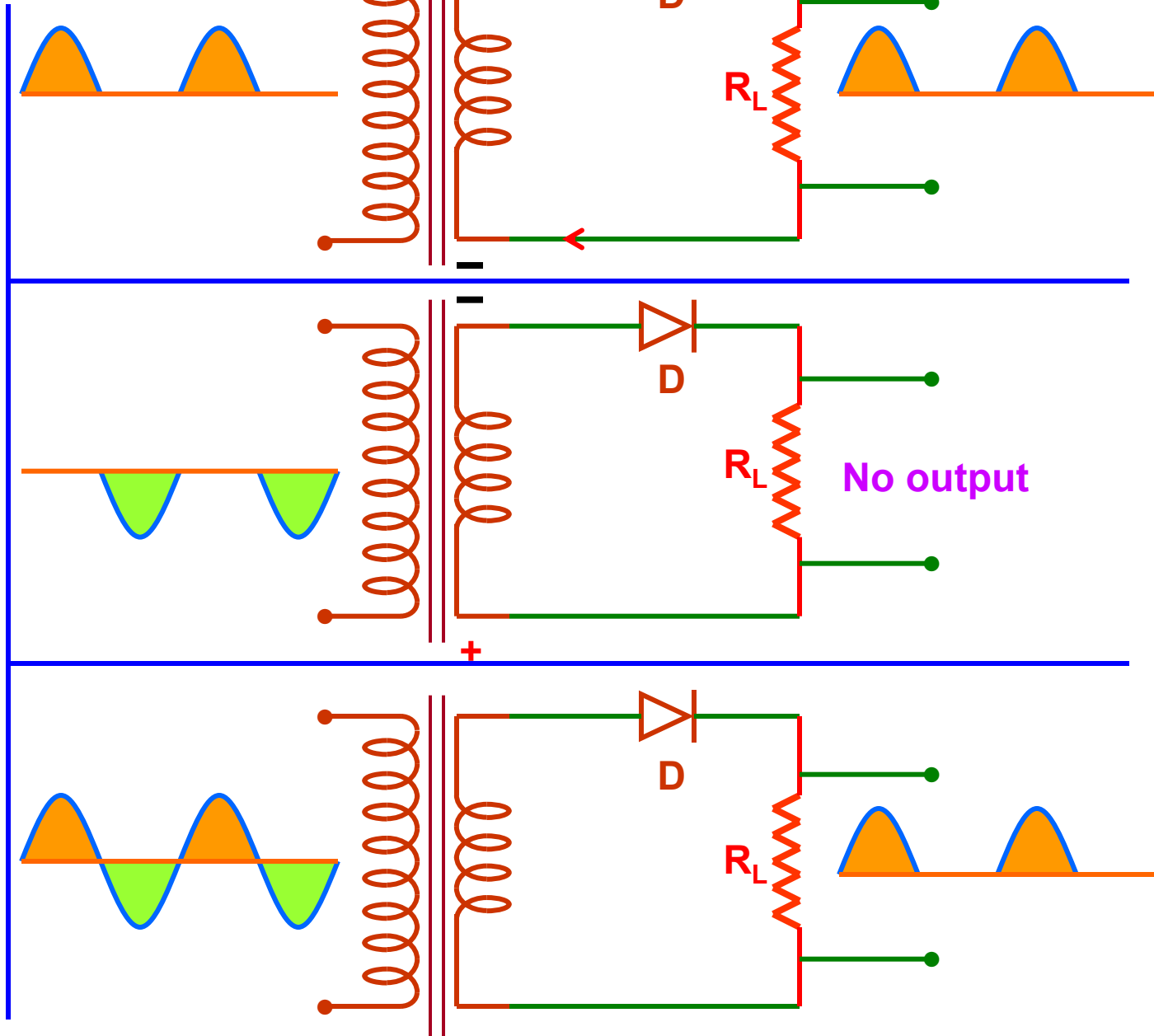
$$R_{a.c} = (n)(V_T)/(I)$$

# PN Junction Diode as a Half Wave Rectifier:

The process of converting alternating current into direct current is called **'rectification'**.

The device used for rectification is called **'rectifier'**.

The PN junction diode offers low resistance in forward bias and high resistance in reverse bias.

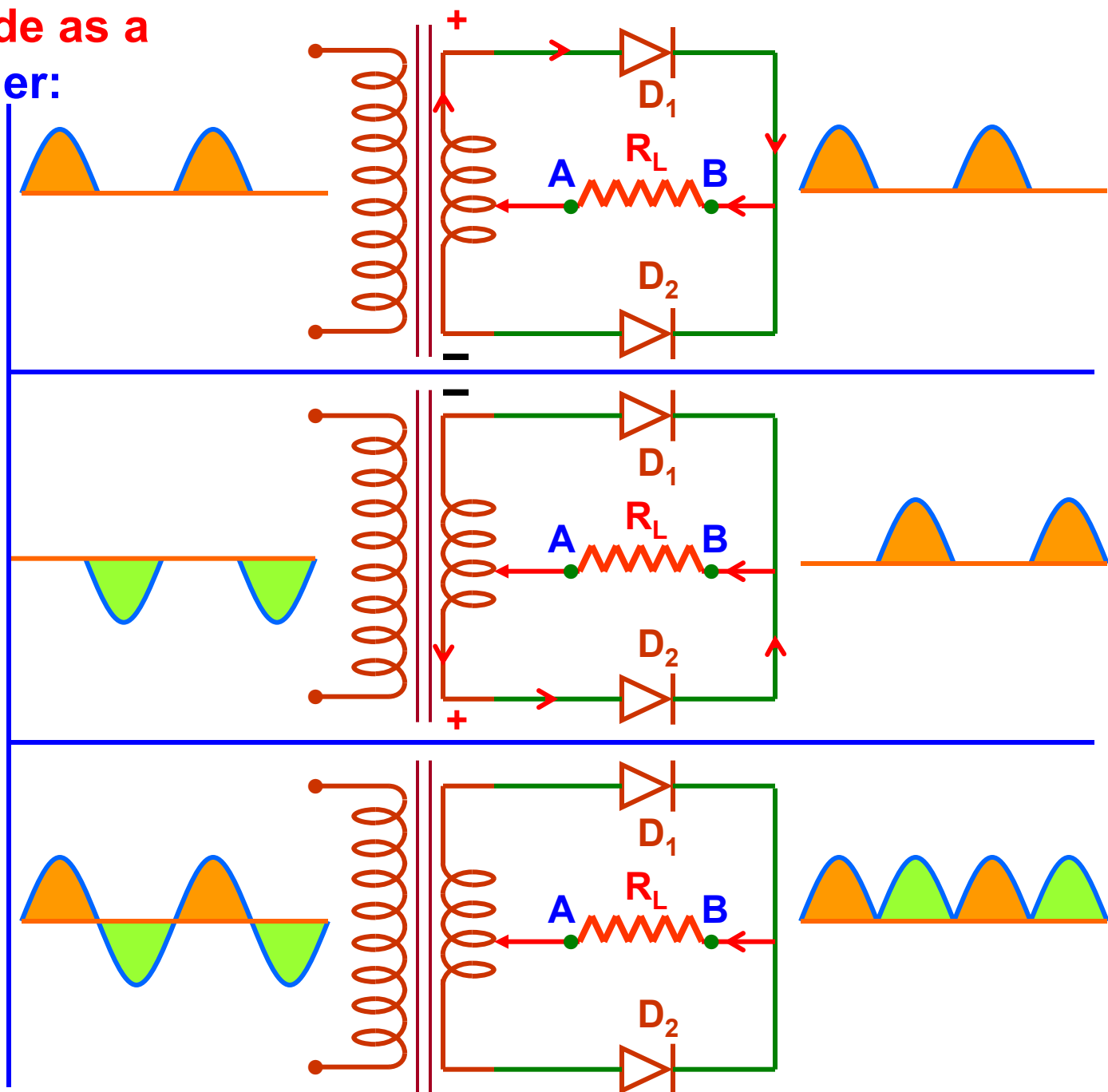


# PN Junction Diode as a Full Wave Rectifier:

When the diode rectifies whole of the AC wave, it is called 'full wave rectifier'.

During the positive half cycle of the input ac signal, the diode  $D_1$  conducts and current is through **BA**.

During the negative half cycle, the diode  $D_2$  conducts and current is through **BA**.



# Half Wave Rectifier

$$\text{Efficiency } (\eta) = \frac{\text{DC Output Power}}{\text{AC Input Power}}$$

$$\text{DC Output Power of Half Wave Rectifier} = V_o \cdot I_o = (I_o)^2 \cdot R_L$$

$$\text{AC Input Power of Half Wave Rectifier} = V_{or} \cdot I_{or} = (I_{or})^2 \cdot R_L$$

$$\text{Efficiency } (\eta) = \frac{V_o \cdot I_o}{V_{or} \cdot I_{or}} = \frac{(I_o)^2 \cdot R_L}{(I_{or})^2 \cdot R_L}$$

Assuming Diode used in this circuit is ideal means its forward resistance ( $R_F$ ) is 0.

$$\text{Efficiency } (\eta) = \frac{(V_m / \pi)^2}{(V_m / 2)^2} = \frac{4}{\pi^2} = 40.5\%$$

# Full Wave Rectifier

$$\eta = \frac{P_{DC \text{ output}}}{P_{AC \text{ input}}} = \frac{\frac{4}{\pi^2} I_m^2 R_L}{\frac{I_m^2 (R_f + R_s + R_L)}{2}} = \frac{8 R_L}{\pi^2 (R_f + R_s + R_L)}$$

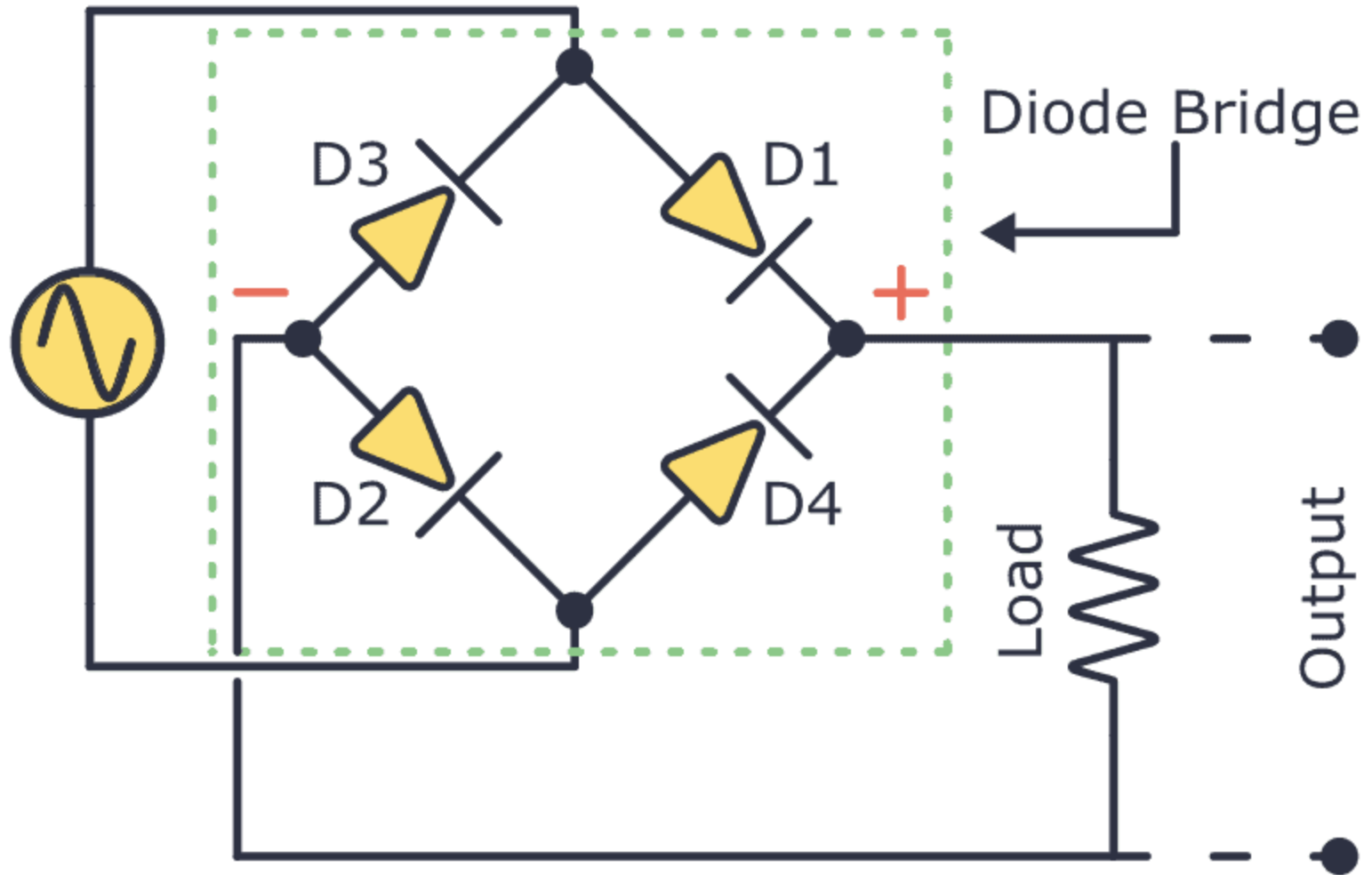
- But if  $R_f + R_s \ll R_L$ , neglecting it from denominator we get maximum  $\eta$  as,

$$\eta_{\max} = \frac{8 R_L}{\pi^2 (R_L)} = \frac{8}{\pi^2}$$

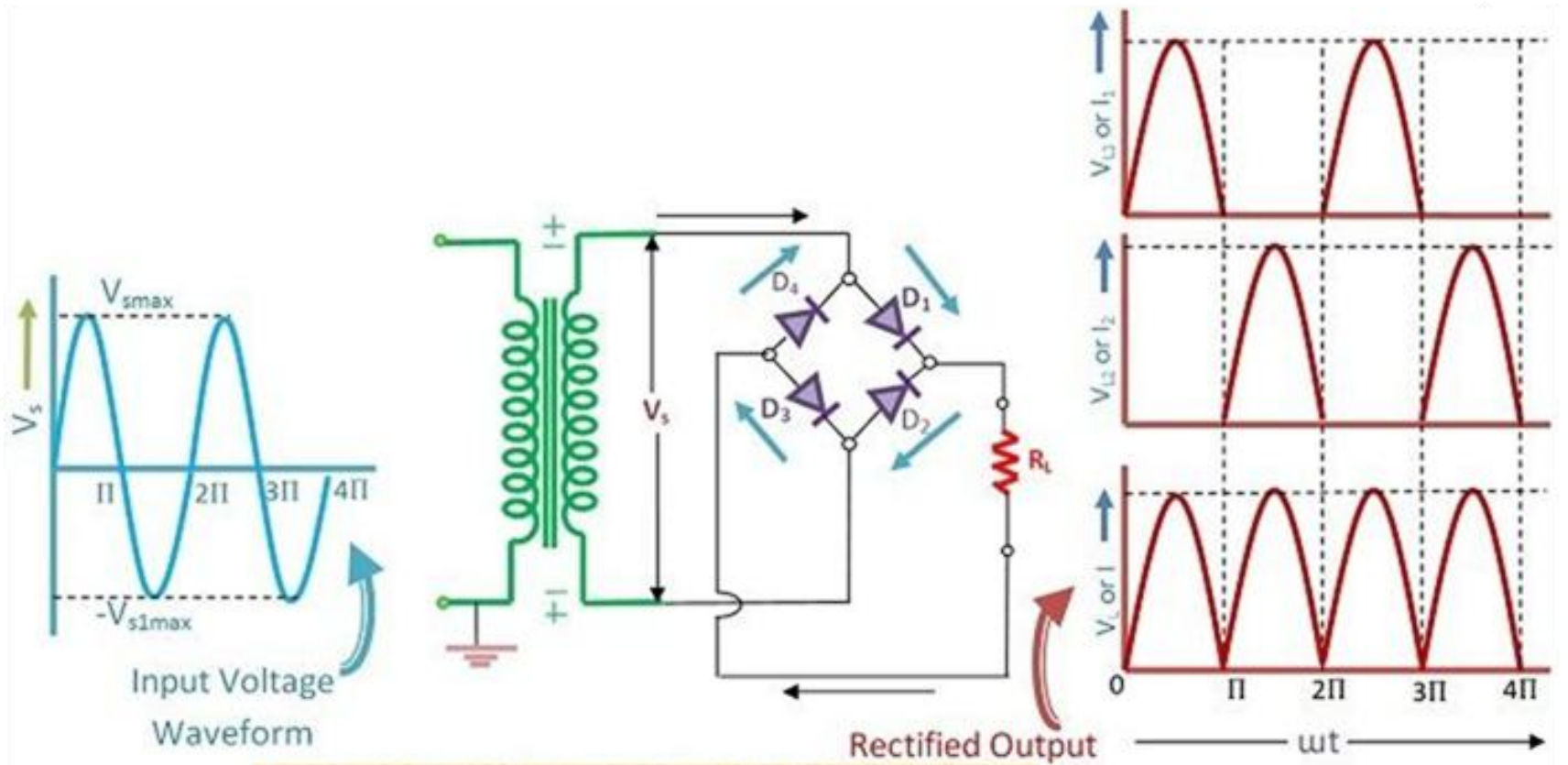
$$\% \eta_{\max} = \frac{8}{\pi^2} \times 100 = 81.2 \%$$

.... Maximum efficiency of full wave rectifier.

# PN Junction Diode as a bridge Rectifier



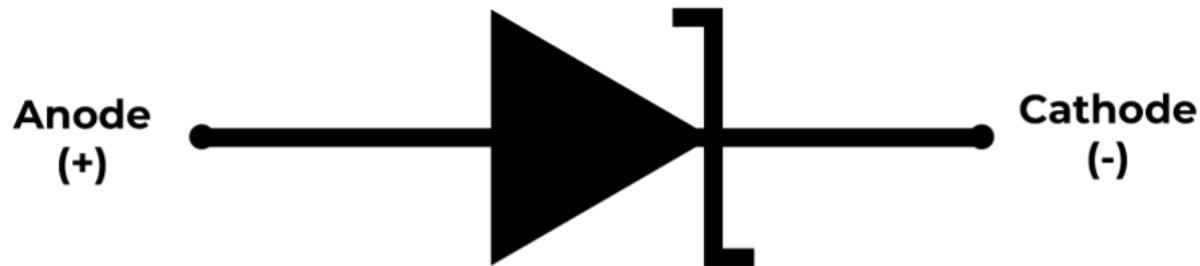




**Bridge Rectifier**

# Zener Diode

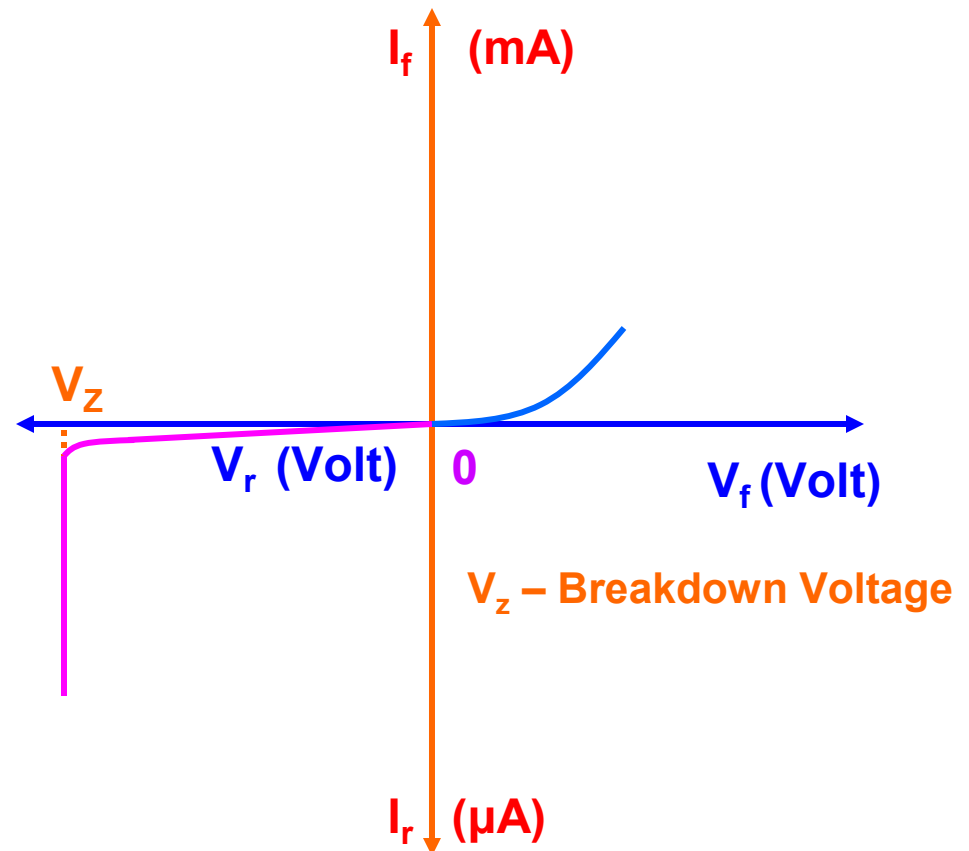
- ❖ A conventional PN junction diode works in forward bias.
- ❖ A special heavily doped p-n junction diode that works in reverse bias conditions is called a Zener Diode or Breakdown diode.
- ❖ For the Zener diode, the voltage drop across the diode is always constant irrespective of the applied voltage.
- ❖ Thus, Zener diodes are used as a voltage regulator.



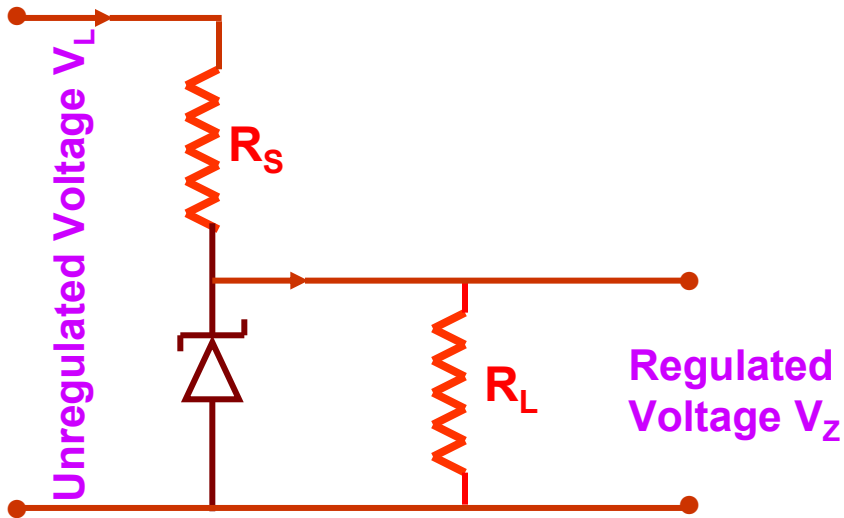
- ❖ In forward-biased conditions, the Zener Diode works like any normal diode but in the reverse-bias condition, a small leak current flows through the diode.
- ❖ As we keep increasing the reverse voltage it reaches a point where the reverse voltage equals the breakdown voltage.

❖ The breakdown voltage is represented as  $V_z$  and in this condition the current start flowing in the diode.

❖ After the breakdown voltage the current increase drastically until it reaches a stable value.



# Zener Diode as a Voltage Regulator



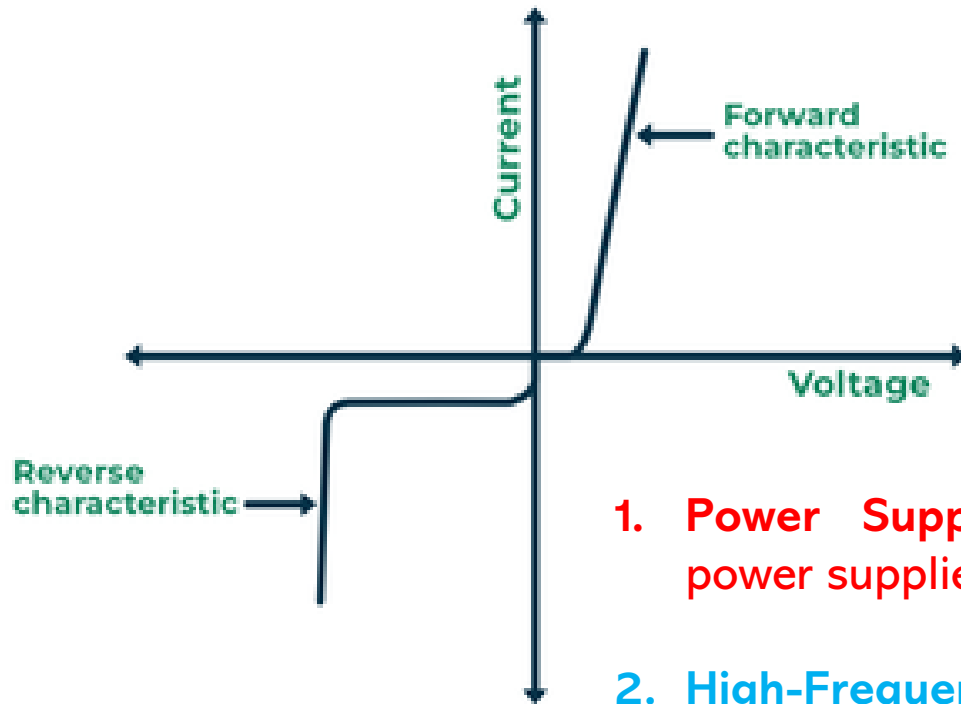
- ❖ A resistor connected in series to the circuit in order to limit the current into the diode.
- ❖ When the minimum input voltage and the maximum load current is applied, the Zener diode current should always be minimum.
- ❖ Since the input voltage and the required output voltage is known, it is easier to choose a Zener diode with a voltage approximately equal to the load voltage, i.e.  $V_Z = V_L$ .
- ❖ Current through the diode increases when the voltage across the diode tends to increase which results in the voltage drop across the resistor.
- ❖ Similarly, the current through the diode decreases when the voltage across the diode tends to decrease.
- ❖ Here, the voltage drop across the resistor is very less, and the output voltage results normally.

# Schottky diode

- ❖ When p-type and n-type semiconductors are joined collectively then it is called PN junction diode.
- ❖ When p-type semiconductor is replaced by metals like aluminum, then it is called Schottky diode.
- ❖ It is a type of diode formed by the junction of a semiconductor with a metal.
- ❖ It is also called as a hot-carrier diode, low-voltage diode, or Schottky barrier diode.
- ❖ It has advantage of low forward bias voltage, large forward current, Efficient for low voltage.



- ❖ Schottky diodes are constructed using semiconductor materials like Silicon (Si).
- ❖ A metal contact (made of metals like Pt, W, Al) is directly connected to the semiconductor material (usually n-type silicon).
- ❖ The region beneath the metal contact is n-type semiconductor. It has a higher concentration of electrons (i.e, electrons are the majority charge carriers).
- ❖ Schottky diodes have a depletion region near the junction. Since there is no p-type region, this depletion region is much thinner than a normal diode.
- ❖ The interaction of electrons from the n-type semiconductor with metal atoms is the major driving factor for formation of this region.



1. **Power Supplies:** Efficient rectifiers, switching power supplies.
2. **High-Frequency Circuits:** Radar detectors, mixers, signal processing.
3. **Logic Gates:** TTL (Transistor-Transistor Logic) for fast digital switching (e.g., 74LS series).
4. **Automotive:** LED protection, alternators for better fuel efficiency